

IMPACT OF THE INTERNET

THE INTERNET

- **Definition:** The internet is the interconnected network of computers that provides many online services to users. The World Wide Web is just one use of the internet.
 - **Protocols:** Rules that allow the exchange and transmission of data between devices.
 - **IMAP (internet message access protocol)** for email.
 - **FTP (file transfer protocol)** for sending files.
 - **SSH (secure shell)** for secure remote logins.
 - **VoIP (voice over internet protocol)**, used for voice call services like Skype.
 - **XMPP (extensible messaging presence protocol)**, used for messaging services like WhatsApp.
 - **HTTP (hypertext transfer protocol)**, used for the World Wide Web.
 - **Key Point:** The internet is connected devices. The World Wide Web is connected content.
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IMPACT OF THE INTERNET ON INDIVIDUALS

INFORMATION AND SERVICES

- The internet improves users with access to information from news, sport and weather services. Information can spread on the internet within seconds.
- Users can also use the internet to access booking systems for travel, leisure and entertainment; do their shopping and banking; study using virtual learning environments (VLEs).
- **Web 2.0:** The development of online interactivity, which allowed users not only to read webpages but also to add information to them.
- **Meme:** A photo, a piece of video, a joke or similar content that spreads quickly on the internet.

EMPLOYMENT

- **Skill Requirements:** Many employers now require employees to use the internet for work, meaning that employees need to learn new skills.
- **New Job Opportunities:** As the internet changes the work that people do, this provides new job opportunities. Skilled workers are needed to train and advise unskilled workers, create digital devices and software, and install and maintain systems.

- **Job Losses:** If employees do not keep updating their skills in the use of the internet, they may not have the skills required to carry out new work. If they are not willing or able to retrain, they could lose their jobs.

WORKING PRACTICES

- **Collaborative Working:** Working together. The internet enables people to connect with one another in real-time, making the process of sharing work more efficient.
 - *Split into individual tasks benefits:* Each employee can focus on one task and become experts through (narrow) experience or training.
 - *Split into individual tasks drawbacks:* Employees do not share skills and have reduced understanding of the whole project.
 - *More than one person on a task benefits:* Expertise is shared and employees can check each other's work.
 - *More than one person on a task drawbacks:* Difficult for lots of people to agree and difficult to co-ordinate the work.
- **Flexible Working:** A way of working that suits an employee's needs, allowing employees to decide the hours that they work or enables them to work from home.
 - *Benefits:* Employees can work at a time right for them, take breaks whenever needed, and fit work around family life.
 - *Drawbacks:* Employees do not meet face to face as much, may not manage work effectively, or cause exhaustion.
- **Mobile Working:** Allows employees to work when they are travelling or on the move to access up-to-date data when on-site.
 - *Benefits:* Access up-to-date information and work from anywhere with an internet connection.
 - *Drawbacks:* Employees could become exhausted if they work longer hours or may not have access to the same facilities.

SOCIAL IMPACT

- **Social Interaction:** Some people are concerned that our increasing use of virtual environments is replacing face-to-face social interaction. This is damaging people's ability to talk to each other and understand each other's emotions.
- **Cyberbullying:** Using the internet to send text, images or multimedia in order to upset or embarrass someone. Information can spread very quickly and to more people, and bullies often hide their identity.
- **Physical Activity:** Online goods and services have reduced the need to travel, and this reduced need for physical activity as part of daily life could have negative effects on people's health and wellbeing.

STAYING SAFE ONLINE

- **Zip It:** Keep your personal information private and do not share it with strangers (location, school name, phone number, real name, photos, mailing or email address).
- **Block It:** Always block offensive messages or friend requests from strangers on social networking sites, and do not open suspicious attachments or links.

- **Flag It:** You should always tell an adult if something online upsets you or if someone you don't know requests to meet you.

DIGITAL FOOTPRINT

- **Definition:** All the data that a user creates online, which is recorded and stored.
 - Much of what you share online can be recorded forever. Content can be easily viewed by people whom you did not expect to see it, copied, and shared easily.
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ONLINE WORKING FROM HOME

- **Benefits to Individuals:** No need to travel to work (reduces stress, saves money, reduces travel emissions), can spend more time working or resting, no need to wear a uniform, can work in a comfortable environment, and organize work around family commitments.
 - **Benefits to Organisations:** Can attract a talented, motivated workforce, employ people who are located anywhere, do not need to buy or rent office space or furniture, and workers will not be delayed by weather or transport issues.
 - **Drawbacks to Individuals:** May be distractions at home, suffer from a lack of social interaction with colleagues, and feel disconnected from the company.
 - **Drawbacks to Organisations:** Concerns about data security, more difficult to manage and support employees, employees might not work as hard, and complicated to organise payments in different countries.
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IMPACT OF THE INTERNET ON ORGANISATIONS

POSITIVE IMPACTS

- **Improved Communication:** Communicate and interact with customers and employees more easily and in real-time using email, instant messaging and social media.
- **Access to Global Markets:** Made it easier for organisations to advertise and sell to customers in countries around the world without physical packaging or postage charges.
- **Access to Global Workforce:** Can employ people in other countries, make use of different skills and time zones to cover all 24 hours in one day, and reduce costs.
- **How Information is Managed and Used:** Organisations can make use of **big data** (the analysis of huge quantities of data collected from many sources) in order to find patterns of behaviour and interactions.

NEGATIVE IMPACTS

- **Security Issues:** Remote access means that data is more vulnerable to being accessed by an unauthorised user. Unauthorised users gain access by: authorised users revealing login details, intercepting data transferred over the internet, or hacking into systems.
 - **Greater Competition:** Organisations must now compete in a global marketplace. Entire industries have been put at risk: music and film retailers, DVD rental companies, record labels, estate agents, newspaper publishers, and postal services.
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IMPACT OF THE INTERNET ON SOCIETY

THE CAUSES OF UNEQUAL ACCESS TO ICT

- **Digital Divide:** The gap between people who have access to digital devices and the internet and people who do not. Gaps are caused by:
 - *Economy and infrastructure:* Developing economies prioritise other needs like food and healthcare.
 - *Location:* Remote, rural, heavily forested and mountainous areas where wireless signals cannot reach.
 - *Politics:* Governments deliberately block, restrict, or disrupt internet access to control their citizens.
 - *Religion, Disability, and Social Factors:* Age, gender, education, and income affect access.

THE GAP BETWEEN INFORMATION RICH AND INFORMATION POOR

- **Information Rich:** People with good access to information, which improves their level of education and enhances their ability to make decisions.
- **Information Poor:** People with limited access to information, which decreases their education and reduces their ability to make decisions.

THE IMPACT OF LESS RESTRICTED ACCESS TO NETWORKS

- Social media and video streaming services have enabled people to broadcast their own messages and interact with global audiences.
- User-generated reference sites like Wikipedia have challenged society's ideas about expertise because truth, knowledge and accuracy are no longer checked by a limited number of experts.
- Networks have had an impact on law enforcement both negatively (rioters organising themselves via mobile networks) and positively (police using CCTV and smartphone footage to identify those responsible).

CHANGES IN WAYS OF SOCIALISING

- The internet has arguably affected the way that people think and their ability to form relationships with others.

- It can cause people to become isolated from their own families and local communities because they spend most of their time looking at the screens of their digital devices.
 - Having status updates sent directly means you no longer have to ask people what they are doing, leading some to argue that social media makes friendship meaningless.
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CHAPTER QUESTIONS

1. Which one of these activities is only possible when using the internet? (1)
 - a. Television
 - b. Radio
 - c. Streaming
 - d. Mobile telephone calls
 2. State two benefits to employees of working from home. (2)
 3. State two drawbacks to organisations of employees working from home. (2)
 4. Explain one reason why the use of the internet may affect the ability of individuals to interact with other people face-to-face. (3)
 5. Explain one reason why cyberbullying can be more distressing than in-person bullying. (2)
 6. State what is meant by the term 'collaborative working'. (1)
 7. State three factors that affect unequal access to ICT. (3)
 8. State three examples of services that are not available to those with limited or no access to digital technologies. (3)
 9. Describe two ways in which greater access to networks has affected society. (4)
 10. Describe the effect of the internet on the information gap. (2)
 11. Discuss the impact of the internet on how individuals socialise. (8)
 12. Describe two ways in which individuals can stay safe online. (4)
 13. Explain why organisations' use of the internet increases the risk to data security. (3)
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